❝ ONE **SMALL STEP** FOR [A] **MAN,** ONE **GIANT LEAP** FOR **MANKIND.** ❞

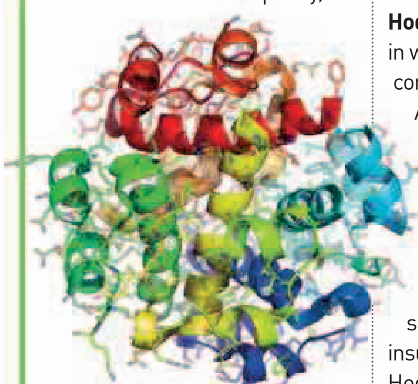
Neil Armstrong, American astronaut, 1969

IN THE WAKE OF Christiaan Barnard's ground-breaking heart transplant (see 1967), American surgeon **Denton Cooley** (b.1920) **implanted the first artificial heart** on April 4. At times when natural hearts were unavailable and surgical intervention urgent, early artificial hearts could give the patient time until a donor could be found. The first patient to receive an artificial heart survived long enough to get a donor.

In an event that was televised live across the world, American astronauts **Neil Armstrong** (1930–2012) and **Buzz Aldrin** (b.1930) became **the first humans to set foot on the Moon** on July 21 Coordinated Universal Time (UTC). Their spacecraft, Apollo 11, had touched down on the Moon's Sea of Tranquillity, a large plain, the day before. Armstrong and Aldrin spent two and a half hours on the surface of the Moon, collecting samples of lunar rock. They left behind instruments, including a series of reflectors for laser-ranging experiments, which would help **determine the distance between Earth and the Moon** to a degree of accuracy never before possible. The astronauts returned to Earth on July 24, splashing down in a module in the Pacific Ocean. The entire mission took just over eight days to complete.

British biochemist **Dorothy Hodgkin** (see 1945) specialized in working out the structures of complex biological molecules. After her success with steroids, penicillin, and vitamins, she moved on to a **much more complicated substance—insulin**, a protein hormone. Fred Sanger had determined the sequential building blocks of insulin in 1951. Ten years later, Hodgkin worked out the **3-D structure of insulin** by using X-ray diffraction techniques that had earlier been applied to DNA and other complex structures.

First man on the Moon
Neil Armstrong was the first person to step on the Moon, followed soon afterward by Buzz Aldrin. Television images of the event were sent back to at least 600 million people on Earth.



Structure of insulin
Dorothy Hodgkin showed how the building blocks of insulin were spatially arranged to form a fixed complex shape.

January Peptide **hormones described** in the hypothalamus of the brain

March Norwegian-American meteorologist Jacob Bjerknes describes **pattern of global climate oscillation,** exemplified by El Niño

March 31 Formaldehyde discovered between stars—the first large **organic molecule discovered in interstellar space**

April 4 Denton Cooley implants the **first temporary artificial heart**

May 16–17, 1969 USSR's Venera 5 and 6 land on Venus

July 20 US spacecraft **Apollo 11** lands on the Moon

July 21 Apollo 11 astronauts Neil Armstrong and Buzz Aldrin become the **first humans to walk on the surface of the Moon**

July American paleontologist John Ostrom points out the **birdlike characteristics** of *Deinonychus*, a **bipedal dinosaur**

August American microbiologists Thomas Brock and Hudson Freeze describe **discovery of bacteria living in hot springs**

November 1 Dorothy Crowfoot (later Hodgkin) publishes **3-D structure of insulin**